Ekkati Laxman Reddy

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EDUCATION

Lovely Professional University, Phagwara, Punjab – *B.Tech, Computer Science & Engineering* | *CGPA: 8.1* Aug 2022 – Present

Relevant Coursework: Natural Language Processing, Machine Learning, Deep Learning, Transformers & LLMs, Distributed Systems, Data Structures & Algorithms, Database Management Systems

Sri Chaitanya Junior College, Hyderabad – *Intermediate* | *96.8%* Apr 2021 – May 2022

INTERNSHIP

Programmer Analyst Trainee – *Cognizant Technology Solutions* Jan 2026 – Present

- Built and delivered production-quality backend services using Java and Python — integrating REST APIs, implementing structured exception handling, and writing testable, well-documented code across enterprise-grade workflows in an Agile sprint environment.
- Collaborated with senior engineers on system design, service orchestration, and iterative feature delivery — contributing to debugging, performance analysis, and operational support cycles evaluated by BU Subject Matter Experts against real production requirements.

SKILLS

Agentic AI:	Multi-agent workflows, tool execution, task orchestration, agent collaboration patterns, LangChain, LlamaIndex, CrewAI (familiar)
RAG & Retrieval:	RAG pipelines, embedding-based search, semantic retrieval, grounding strategies, ChromaDB (familiar), Pinecone (familiar), retrieval quality evaluation
LLM & NLP:	Groq API, OpenAI API (familiar), Prompt Engineering (Zero-shot, Few-shot, CoT), Transformers, NLP pipelines, fine-tuning (familiar), Scikit-learn
Languages:	Python (primary), Java, SQL, Kotlin | NumPy, Pandas, Matplotlib, REST APIs, GIT/GitHub
Infra & Cloud:	Docker, Jenkins CI/CD, Maven, AWS/GCP/Azure (fundamentals), Hugging Face Spaces (deployed), Gradio, Linux

PROJECTS

DataGroq – Agentic RAG System for Document Intelligence *Groq API, LangChain, Python, RAG, NLP, Gradio, HuggingFace* | *Sept 2025* *[Live]*

- Designed and deployed an agentic RAG system enabling natural language querying over **200+ documents** — implementing tool execution, multi-step reasoning, and context synthesis with Groq LLM API to automate complex analytical workflows with non-deterministic LLM behavior handled through structured guardrails.
- Built end-to-end knowledge retrieval pipelines — tokenization, lemmatization, semantic classification, keyword extraction, and embedding-based retrieval — integrating grounding strategies to improve retrieval quality and LLM response accuracy by **30%**.
- Implemented agent collaboration patterns with safe Python code execution, EDA-driven metric extraction, and output validation — reducing manual analysis time by **70%** while maintaining reliable, testable behavior across varied document inputs and query types.

Sentiment Strategy Comparison – AI Evaluation & Regression Testing Framework *Python, TextBlob, Naive Bayes, Transformers, Prompt Engineering, Gradio* | *Dec 2024*

- Built a structured AI evaluation and regression testing framework benchmarking **6 NLP strategies** — TextBlob, Naive Bayes, Zero-shot, One-shot, Few-shot, and Chain-of-Thought — on **1,000+ labeled reviews**, systematically tracking accuracy, latency, and failure mode patterns across model iterations.
- Designed behavior monitoring across prompt strategies — identifying non-deterministic failure modes in generative approaches (avg. **41% accuracy**) vs classical ML (**96.8%**) and generating documented regression insights to guide prompt architecture improvements and model selection decisions.
- Deployed an interactive Gradio experimentation interface for real-time strategy evaluation — enabling rapid prototyping of new agent tool configurations and validation of LLM output quality across varied input conditions aligned with production AI system testing requirements.

Traffic Analysis Dashboard – Multi-API Backend Integration Pipeline *Python, Geoapify APIs, Google Sheets API, Tableau* | *Jan–Feb 2025*

- Engineered a backend data integration pipeline connecting multiple external REST APIs (Geoapify, Google Sheets) — applying Python-based extraction, transformation, and structured validation to deliver reliable, automated data ingestion **reducing manual effort by 80%**.
- Implemented fault-tolerant API orchestration with error recovery, rate limit handling, and anomaly detection across heterogeneous data sources — contributing to a **15% gain in operational efficiency** through resilient, self-healing pipeline execution.
- Delivered a dynamic Tableau dashboard providing real-time geospatial analytics and operational KPI visibility — enabling data-driven workflow automation decisions that directly mirror TraceLink's mission of improving supply chain visibility and operational intelligence.

CERTIFICATIONS & ACHIEVEMENTS

Introduction to Machine Learning – NPTEL (Oct 2024) | **Data Visualization with Tableau** – Coursera (Nov 2024) | **Python, Java & SQL** – HackerRank (Jan 2025)

Achieved **5-Star rating in SQL** on HackerRank (Feb 2025) | Active Open Source Contributor & Hackathon Participant